

# Single-Cell ML Feature Engineering Reference

## Data Export Specification & Downstream Modeling Considerations

This document provides a concise engineering overview of the two master datasets exported from the single-cell RNA-sequencing (scRNA-seq) pipeline. These matrices are structured to benchmark different analytical approaches for predicting patient clinical severity using machine learning.

### 1. The Latent Space Matrix

File Name: `scvi_latent_representation_full.parquet`

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| Dimensions    | Rows: All single cells (unfiltered cohort) × Columns: 26                                  |
| Core Features | 20 continuous numerical variables ( <code>scVI_1</code> to <code>scVI_20</code> )         |
| Metadata Keys | Cell type, WHO score (0–8), Participant ID, Variant Group, Vaccination Status, PCR Status |

#### Biological & Machine Learning Profile:

- **The Global Cellular Landscape:** Represents a dense, continuous, low-dimensional coordinate mapping of the complete biological manifold. This space mathematically integrates cell identity and patient background while stripping technical batch effects.
- **All-Inclusive Composition:** Retains 100% of captured cell distributions, preserving healthy controls (WHO 0) and intermediate clinical transitions (WHO 4) omitted during differential expression testing.
- **Interpretability Trade-off:** High predictive utility but low direct biological explanation. Columns represent abstract deep-learning latent dimensions rather than concrete physical biomarkers.

### 2. The DE Gene Expression Matrix

File Name: `de_genes_expression_matrix.parquet`

|               |                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------|
| Dimensions    | Rows: All single cells (identical index alignment) × Columns: 231                                      |
| Core Features | 225 unique, relaxed-threshold differentially expressed genes (e.g., Macrophage & Dendritic signatures) |
| Metadata Keys | Identical 6-column metadata tracking structure for perfect spatial alignment                           |

#### Biological & Machine Learning Profile:

- **Targeted Molecular Signatures:** Captures direct transcriptomic levers driving disease divergence within the hyper-inflammatory myeloid compartment (Macrophages and Dendritic cells).

- **100% Interpretability:** Features map directly to established molecular pathways, allowing direct identification of the precise biomarkers driving the machine learning model's output.
- **Cell-Type Specific Noise:** Because these 225 genes were prioritized from myeloid signatures, their expression matrices will structurally display high zero-inflation (dropout) or static baseline noise when evaluated on lymphoid lineages (e.g., T or B cells).

### 3. Critical Architectural Engineering Rules

#### **Rule 1: Prevent Patient-Level Data Leakage**

Clinical severity ( `WHO_score` ) is a patient-level metric, but the matrices are structured at single-cell resolution. Splitting rows randomly will leak identical patient genetic backgrounds into both training and validation sets, yielding highly artificial accuracy inflation. **Always enforce group-based cross-validation splits utilizing the `Participant` key (e.g., `GroupKFold`).**

#### **Rule 2: Implement Cellular Aggregation (Pseudobulk)**

An individual cell cannot manifest systemic clinical severity. To model patient outcomes validly, aggregate cell rows up to the patient or patient-celltype level (e.g., computing the mathematical mean or variance of features grouped by `Participant` ) before training classification or regression models.

#### **Rule 3: Re-align Target Ranges for Regression**

While intermediate buffers (WHO 4) and healthy controls (WHO 0) were explicitly stripped from the differential expression step to optimize contrast, they remain fully intact within these master files. Ensure downstream ordinal or continuous loss functions are configured to receive the full continuous spectrum from 0 to 8.